

Trajectory-resolved adaptive preparation of a charge-fluctuating chiral critical ensemble

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This note summarizes the present numerical evidence that the finite-range class-A adaptive fermion circuit prepares a charge-fluctuating, quasi-one-dimensional chiral critical ensemble localized at a Chern-insulator domain wall. The central distinction is ensemble-theoretic. The physically meaningful ideal protocol is trajectory-resolved perfect correction: Born measurement outcomes are sampled, ideal feedback is applied, and observables are averaged over the resulting random Gaussian trajectories. Post-selection, the deterministic Markov channel, and its continuous-time Lindblad limit remove this randomness in inequivalent ways and should be regarded as controls. The existing data show robust logarithmic entanglement scaling near the chiral-fermion value, purification toward a topological domain-wall ensemble, persistent charge-sector fluctuations under perfect correction, strong suppression of such fluctuations under post-selection, and modular-charge spreading consistent with opposite chirality on opposite walls. The evidence is coherent enough to support the paper narrative, but not yet sufficient to close all finite-size and correlator universality questions.

I. PURPOSE AND RELATION TO THE PREVIOUS WORK

Ref. [1] developed the general framework: fermionic dynamics with measurements can be classified by many-body evolution-operator and single-particle transfer matrix symmetry classes, and class-A Gaussian adaptive circuits can prepare and stabilize Chern-insulator states without post-selection. It also identified dynamical domain-wall modes and a separation between trajectory-resolved and trajectory-averaged transitions.

The present note is narrower and more diagnostic. It asks whether the finite-range class-A adaptive circuit, in a domain-wall geometry, prepares not merely a topological bulk slab but a quasi-one-dimensional chiral critical state ensemble on the wall. The answer suggested by the current data is affirmative, provided that one keeps the ensemble structure explicit. Perfect correction is not equivalent to post-selection, and neither is equivalent to evolving the averaged covariance matrix. This difference is not a technical detail: it is the mechanism behind the observed charge-fluctuating phenomenology.

II. FOUR DYNAMICAL OBJECTS

The top-layer Gaussian state in a trajectory is encoded by a covariance matrix G_ξ , or equivalently by the one-body density matrix

$$C_\xi = \frac{1 + G_\xi}{2}. \quad (1)$$

The measurement record ξ labels the sequence of Born outcomes. A nonlinear observable O defines at least three inequivalent quantities:

$$\overline{O[G_\xi(t)]} \neq O[\overline{G_\xi(t)}] \neq O[G_{ps}(t)]. \quad (2)$$

Figure 1 summarizes the taxonomy.

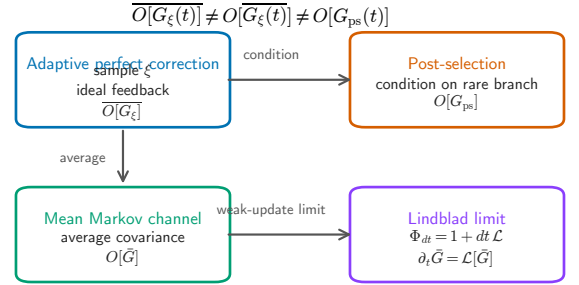


FIG. 1. The four descriptions used in this note. Perfect correction is the trajectory-resolved adaptive ensemble. Post-selection is a fixed conditioned branch. The Markov channel evolves the mean covariance. The Lindblad equation is the weak-update continuous-time limit of that mean channel.

A. Adaptive perfect correction

In the physical ideal protocol, measurement outcomes are sampled with their Born probabilities. Whenever an outcome disagrees with the target band occupation, ideal feedback applies the corresponding correction. In code language, a failed empty-band condition triggers deterministic pump-out feedback, and a failed filled-band condition triggers deterministic pump-in feedback. The measurement outcome itself is still random. The trajectory remains Gaussian, but the ensemble does not collapse to a single covariance. The natural object is therefore

$$\overline{O[G_\xi]} = \sum_{\xi} p_{\xi} O[G_{\xi}], \quad (3)$$

with p_{ξ} determined dynamically by the Born rule.

B. Post-selection

Post-selection follows only the branch in which every measurement outcome is already the desired one. It computes $O[G_{\text{ps}}]$, where G_{ps} is a conditioned covariance. This is a useful reference because it exposes what the adaptive circuit would do if measurement-record randomness were removed rather than corrected. The numerical driver correspondingly forces the post-selected run to a single branch. It is not a physical preparation protocol: the selected branch has exponentially small weight in system size and time in generic monitored dynamics.

C. Mean Markov channel

The deterministic routine `run_markov_channel` evolves a single covariance by the Born-rule averaged local channel. It gives a discrete map

$$\overline{G}_{\xi_{n+1}} = \Phi(\overline{G}_{\xi_n}), \quad (4)$$

and therefore computes $O[\overline{G}_{\xi}]$. This is distinct from $O[G_{\xi}]$ whenever O is nonlinear. Entanglement entropies, entanglement contours, Chern markers built from regularized spectra, and squared correlation functions all fall into this nonlinear class. The existing correlation-profile comparison in Fig. 4 makes the distinction visible directly.

D. Lindblad limit

The continuous-time Lindbladian construction in Ref. [1] and in the related internal Lindblad note is the weak-success-probability limit of the mean channel. For a local elementary channel Φ_{dt} ,

$$\Phi_{dt} = 1 + dt \mathcal{L} + O(dt^2), \quad \partial_t \overline{G}_{\xi} = \mathcal{L}[\overline{G}_{\xi}]. \quad (5)$$

This limit is analytically valuable: it identifies the jump-process structure of the trajectory-averaged preparation and explains convergence toward the Chern-insulator covariance. Discrete data at update probability $p = 1$ are Markov-channel data, not by themselves a controlled continuous-time extrapolation. The Lindblad limit should therefore not be interpreted as a substitute for trajectory-resolved adaptive dynamics.

III. TARGET DOMAIN-WALL CRITICAL THEORY

The target fixed point is the interface between a class-A Chern slab and a trivial exterior. In the ideal continuum description the wall carries a chiral complex fermion with central charge $c = 1$. For an interval of length ℓ on

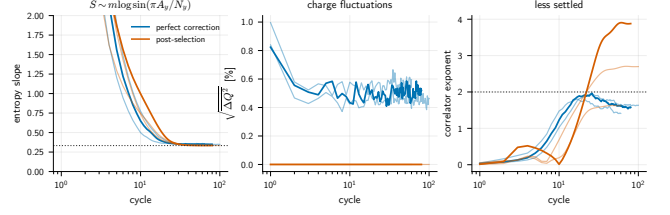


FIG. 2. Init-default streaming diagnostics. Entanglement slopes flow toward the chiral value $1/3$, while the charge-sector RMS separates the physical stochastic perfect-correction ensemble from the post-selected branch. Correlator exponents are shown as a warning: their finite-size and protocol dependence is stronger than the entropy-slope evidence.

a periodic edge of length L_y , the entanglement contribution scales as

$$S(\ell) = \frac{c}{3} \log \left[\frac{L_y}{\pi a} \sin \left(\frac{\pi \ell}{L_y} \right) \right] + s_0 \quad (c = 1), \quad (6)$$

when the full transverse support of both domain walls is included. The squared equal-time fermion correlation along the wall is expected to decay as

$$|C(y) - C(0)|^2 \sim \left[\frac{L_y}{\pi} \sin \left(\frac{\pi |y|}{L_y} \right) \right]^{-2}. \quad (7)$$

The exact flattened-Hamiltonian domain-wall benchmark supplies the calibration for these two targets. The adaptive-circuit data should be judged against Eqs. (6) and (7), but with finite-size and ensemble-ordering caveats.

IV. INIT-DEFAULT ADAPTIVE DYNAMICS

The init-default runs show rapid flow toward the entanglement-scaling target. Figure 2 compares perfect correction and post-selection across the selected $N_x = 20$ geometries. The late-time entropy slopes under both protocols approach values near $1/3$, but the charge-sector behavior is sharply different: perfect correction retains trajectory-to-trajectory charge fluctuations, while post-selection suppresses them to machine precision in the available data.

The key interpretation is not that post-selection and perfect correction are numerically identical. Rather, the entropy slope is a robust chiral-critical diagnostic, while the charge-sector statistics remember whether measurement randomness was sampled or removed. The perfect-correction ensemble is therefore the object relevant to physical adaptive preparation.

V. MAX-MIX PURIFICATION

The max-mix purification campaigns provide a complementary view. Starting from a highly mixed covari-

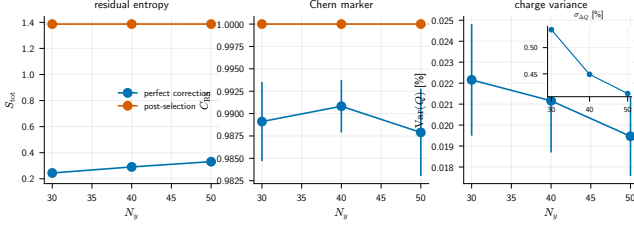


FIG. 3. Max-mix purification at final cycle for the $N_x = 20$ campaign. Perfect correction and post-selection both recover the topological marker, but they differ strongly in residual entropy and charge fluctuations. The dotted entropy line marks $2 \log 2$.

ance, the adaptive protocol purifies toward a topological domain-wall ensemble. Figure 3 shows that perfect correction drives the residual entropy well below the post-selected value, while preserving small but nonzero charge fluctuations. Post-selection gives a cleaner Chern marker and a characteristic residual entropy near $2 \log 2$, but it does so by following the conditioned branch.

This distinction is precisely the desired phenomenology. The protocol is not merely projecting onto a fixed charge sector. Under perfect correction the measurement record remains random and generates a charge-fluctuating ensemble of chiral critical states.

VI. TRAJECTORY-RESOLVED OBSERVABLES VERSUS THE MEAN CHANNEL

The sharpest mathematical point is the noncommutativity of nonlinear observables with trajectory averaging. For a linear one-body observable A ,

$$\overline{\text{Tr}(AC_\xi)} = \text{Tr}(A\bar{C}). \quad (8)$$

For entropy and squared correlations this simplification fails. For example,

$$|\overline{C_\xi(r, r')}|^2 = |\bar{C}(r, r')|^2 + \overline{|C_\xi(r, r') - \bar{C}(r, r')|^2}. \quad (9)$$

The second term is exactly the trajectory-to-trajectory fluctuation contribution. Removing it by studying only $|\bar{C}|^2$ changes the long-distance diagnostic.

The Lindblad approximation inherits this status. It is a continuous-time description of the mean channel, not of the full stochastic ensemble. It is therefore appropriate for explaining averaged covariance convergence, but not for replacing the trajectory-resolved critical-state diagnostics.

VII. ENTANGLEMENT AND CORRELATION SCALING

Figure 5 collects the main scaling evidence. The large- N_y entanglement fits approach slopes in the vicinity of

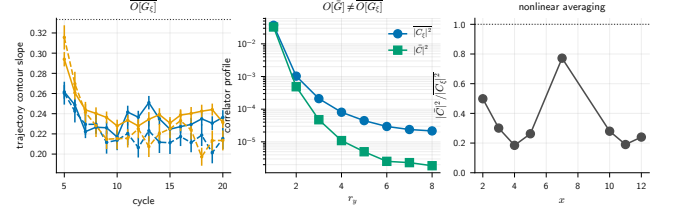


FIG. 4. Trajectory-resolved adaptive diagnostics compared with mean-covariance controls. Left: trajectory-resolved contour-slope statistics from sampled perfect-correction histories. Center and right: existing correlation-profile data showing $|\overline{C_\xi}|^2$ differs systematically from $|\bar{C}|^2$.

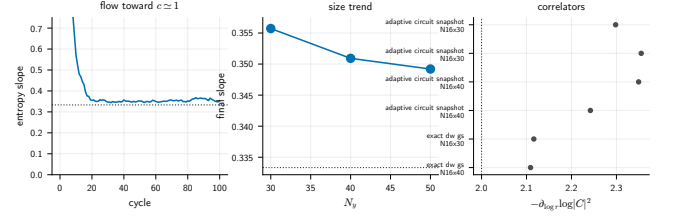


FIG. 5. Critical-scaling summary. Entanglement slopes approach the chiral-fermion target more cleanly than the currently available squared-correlation exponents. The correlator panel should be read as evidence for consistency with chiral scaling, not as a final universality extraction.

$1/3$ with high linear-fit quality. Across $N_y = 30, 40, 50$ final-cycle runs the slopes sit slightly above the ideal value, consistent with finite-size corrections and finite circuit depth. The squared-correlation fits are qualitatively compatible with Eq. (7), but are less settled: fixed- x fits are strongly wall-position dependent, while x -averaged fits are closer to the expected exponent.

At the present stage the responsible claim is therefore asymmetric. The entanglement data strongly support a quasi-one-dimensional chiral critical ensemble. The correlation data support the same picture but require more careful finite-size, wall-window, and trajectory-ordering analysis before being used as a precision exponent measurement.

VIII. MODULAR CHARGE SPREADING AND CHIRALITY

The modular-charge spreading diagnostics probe the orientation of the interface dynamics. The construction inserts a localized modular charge packet and tracks its center of mass under the modular Hamiltonian generated by the covariance. The relevant signatures are the signed displacement

$$\Delta y_{\text{COM}}(t) = y_{\text{COM}}(t) - y_{\text{COM}}(0), \quad (10)$$

and the fitted velocity v_y over a specified time window.

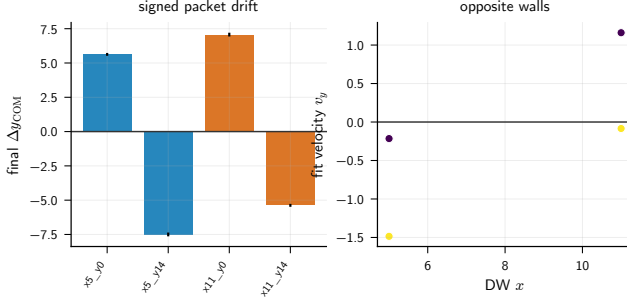


FIG. 6. Modular-charge spreading. Packets launched on opposite domain walls acquire opposite signed drift, and fitted velocities change sign between the walls. This is the most direct current evidence that the domain-wall sector is chiral rather than merely localized.

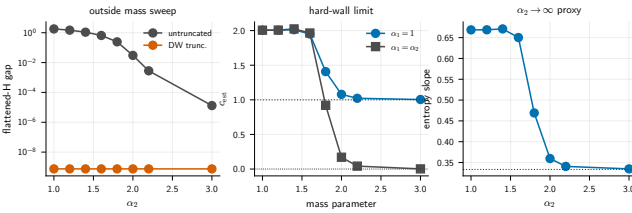


FIG. 7. Flattened-Hamiltonian α_2 sweep. With a topological interior $\alpha_1 = 1$, increasing exterior mass under domain-wall truncation flows toward $c_{\text{est}} \simeq 1$ and entropy slope $1/3$. Uniform-mass truncation alone instead collapses toward $c \simeq 0$ at large mass. Thus truncation is an infinite-mass exterior approximation, not a stand-alone source of chirality.

Figure 6 shows the essential result: packets on opposite walls drift in opposite directions, as expected for a pair of counter-propagating chiral interfaces on a periodic strip. Charge conservation diagnostics in the saved summaries are at numerical precision, so the observed motion is not a charge-leakage artifact.

IX. HARD-WALL INTERPRETATION OF $dw_truncation$

The flattened-Hamiltonian α_2 sweep clarifies how to interpret the production choice `dw_truncation=True`. Unless otherwise stated, the production GPU scaling data use $\alpha_1 = 1$ in the topological slab, a large trivial exterior mass $\alpha_2 = 30$, $n_{\text{shell}} = 1$, and `dw_truncation=True`. In the model, α_1 controls the topological slab and α_2 controls the exterior. Taking α_2 large makes the exterior an inert trivial region. The truncation operation enforces the corresponding support constraint directly at the overcomplete-Wannier projector level: after the finite shell cutoff, each mode is masked to the domain-wall region containing its center and then renormalized.

The correct statement is therefore not that

$$dw_truncation \equiv \alpha_2 = \infty \quad (11)$$

as a literal equality of finite Hamiltonians. Rather, `dw_truncation` implements the hard-wall projector-support limit of an infinite-mass trivial exterior. This distinction matters because uniform-mass truncation by itself does not generate the desired large-mass chiral state; one needs a topological interior adjacent to an inert exterior. The α_2 sweep should therefore be read as a diagnostic proxy for the hard-wall limit, not as a fully controlled extrapolation without further finite-size and truncation checks.

X. LITERATURE POSITION

The broad monitored-circuit context is the measurement-induced entanglement transition literature [2, 3]. For monitored free fermions, there is now a substantial field-theoretic literature on replica/Keldysh and nonlinear-sigma model descriptions [4–8]. Those works are essential background for measurement-induced criticality, but the present question is different: the monitored degrees of freedom are class-A Chern-band projectors, and the critical sector is tied to a domain-wall chiral edge rather than to a one-dimensional bulk monitoring transition.

The distinction between physical Born-rule trajectories and forced or post-selected ensembles is also not optional. Different trajectory ensembles can have different replica limits and different universal behavior [9–11]. This is the conceptual reason that post-selection is treated here as a control, not as the physical adaptive preparation.

The dissipative-control context is the Lindblad preparation of Chern insulators [12–14] and the topology of Gaussian mixed states [15–17]. Those works are closest to the mean-channel and Lindblad-limit controls, not to the full trajectory-resolved perfect-correction ensemble. The monitored topological-dynamics setting is closer to the transfer-matrix and adaptive-circuit programs [1, 18, 19], with the present note zooming in on the chiral domain-wall sector.

Finally, the modular-charge results are naturally compared with chiral modes and transport along Chern-insulator or quantum-anomalous-Hall domain walls [20–24], while remaining a covariance/modular-flow diagnostic rather than a transport measurement in a material.

XI. OPEN NUMERICAL ISSUES

Several points should be settled before this becomes a final manuscript. First, the correlator exponent extraction needs a controlled comparison of fixed-wall, wall-window, and full- x averaged observables, with the ensemble ordering fixed throughout. Second, the Markov-channel and Lindblad controls should be used to explain

which features survive covariance averaging and which are specifically trajectory-resolved. Third, the modular-charge chirality evidence should be extended to more sizes and launch windows. Fourth, the α_2 sweep should be exported into a stable table rather than remaining encoded in notebook outputs. None of these caveats undercuts the main story, but they determine what can be claimed quantitatively.

XII. CONCLUSION

The coherent story is now clear. The finite-range class-A adaptive circuit does more than prepare a bulk Chern marker. Under trajectory-resolved perfect correction it prepares an ensemble whose domain-wall sector has logarithmic entanglement scaling, charge-sector fluctuations, and chiral modular-charge spreading. Post-selection, the

Markov channel, and the Lindblad limit are indispensable controls precisely because they remove or average the measurement-record randomness in different ways. The physical phenomenon is the stochastic adaptive ensemble itself:

$\overline{O[G_\xi]}$ is the object prepared by perfect correction.

It is neither the mean-channel observable $O[\overline{G_\xi}]$ nor the post-selected observable $O[G_{\text{ps}}]$.

ACKNOWLEDGMENTS

This note was generated as a repository-level synthesis of existing data products, figures, and notebooks in `classA_fermionic_adaptive_circuit`. It excludes `erroneous_gpu_stuff`.

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